Global Shocks, Contaminated Controls: Diagnosing Counterfactual Failure in Trump's 2018 and 2025 Tariff Events*

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Abstract

We study what counterfactual methods can and cannot deliver when a policy shock treats every candidate control at once. On Donald Trump's April 2025 "Liberation Day" the S&P 500 fell 12.1% over the four trading sessions from April 3 to April 8 before rebounding on the April 9 pause; every candidate control—U.S. sector ETFs (slices of the treated index) and overseas indices (all sharply down over the same window)—was itself treated. We diagnose the consequences rather than report effects. Placebo tests have no power by construction when donors are homogeneous slices of the treated unit; synthetic weights concentrate on the single best pre-period tracker; and counterfactual gaps order monotonically with how tightly the counterfactual is bound to its donors—near zero under the tightest binding, approaching the raw return as binding loosens. The spread across specifications therefore measures control construction, not treatment effects, and the various "gaps" are different estimands rather than one quantity with uncertainty. A mediation-style attribution of the crash to a VIX "fear channel" fails on two counts: sequential ignorability is violated, and VIX is the implied volatility of S&P 500 options, so the exercise measures mechanical contemporaneous covariation, not a channel. What survives without any donor is descriptive: VIX rose about 140% from its pre-announcement close to its April 8 peak in the 2025 crash and decayed with a one-day half-life, while the comparable 2018 node produced no immediate VIX reaction—a contrast consistent with, but not identified as, market learning. We distill these diagnostics into a checklist for synthetic-control applications to global financial shocks.

Keywords: synthetic control; contaminated controls; event study; tariff shock; identification failure; volatility
JEL: G14, G18, F13, C54

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1 Introduction

The tariff policies of Trump's two presidencies constitute the largest American trade intervention in a century, and they arrived in two delivery mechanisms: the gradual 2018–19 escalation through Section 232 and four Section 301 lists (roughly $370 billion of goods cumulatively), and the shock of April 2, 2025 ("Liberation Day"), when a universal baseline plus reciprocal rates were announced against virtually all trading partners at once. A natural question for market participants and policy evaluators alike: what did these shocks do to the S&P 500 relative to the counterfactual of no tariff war?

This paper's answer is that the question cannot be answered by the standard toolkit in this setting, and that the reasons are instructive. The synthetic control method of Abadie et al. (2010) and its Bayesian-regression cousin (Brodersen et al., 2015) require controls untouched by the intervention. A global tariff shock treats every candidate control simultaneously: U.S. sector ETFs are components of the treated index; every overseas developed market fell sharply over the same event window. Once this is recognized, the interesting research question shifts from "what is the effect?" to "what do these estimators produce when their identifying assumption fails in a known direction, and how can a researcher tell?"

We therefore reframe the exercise as a *failure diagnosis with a surviving descriptive core*. Across two tariff rounds, two donor families, and a battery of estimators (convex SCM, augmented SCM with ridge regularization, BSTS regression, linear and neural-network maps), we document four systematic facts and one set of donor-free findings:

1. **No power by construction.** When donors are homogeneous slices of the treated unit, the synthetic control reconstructs the treated series and the gap tends to zero regardless of the intervention; permutation placebos are then uninformative ($p = 100\%$: the realized gap is the *smallest* in absolute value among all units).

2. **Weight concentration.** In the overseas pool, SCM loads 0.76 on a single index (Canada) and parallel trends fail outright; enlarging the pool does not dilute the concentration.

3. **Gaps order with donor-binding tightness.** Across an augmented-SCM ridge sweep, the gap moves monotonically from the convex bound (near zero) to the raw return itself as regularization loosens the donors' grip on the counterfactual. A statistic that orders with a tuning parameter of the control construction is not measuring the treatment.

4. **Different estimands masquerading as one interval.** The spread of long-window statistics across specifications (roughly two orders of magnitude) is sometimes presented as an "uncertainty boundary." It is better described as a collection of different estimands—S&P 500 minus sector-synthetic, S&P 500 minus an overseas-predicted path, nonlinear prediction errors—none of which is a tariff effect.

What survives without any control group is descriptive: the volatility signature. VIX rose from its eve-of-announcement close of 21.8 (April 1) to 52.3 at the April 8 trough of the 2025 crash (+140%) and decayed with a one-day half-life after the April 9 pause; the comparable 2018 node (List 1) produced no immediate VIX reaction at all. We present this contrast as a descriptive fact and explicitly decline the causal reading ("markets learned") because treatment magnitude, surprise, and macro background all differ across rounds.

A further negative result concerns a tempting exercise: decomposing the crash into a VIX-mediated "fear channel" and a residual direct effect, in the style of Imai et al. (2010). The exercise fails identification twice. Sequential ignorability is violated (VIX responds to the equity decline itself), and—more fundamentally—VIX is the implied volatility of S&P 500 options: regressing index returns on ΔVIX is a mechanical contemporaneous covariation (the leverage/volatility-feedback relation), not the measurement of a channel. We report the decomposition only as a descriptive covariation account, with its fragilities stated.

Contributions.

1. **A diagnosis protocol** for counterfactual methods under globally contaminated controls: constructive no-power placebos, weight-concentration checks, and a binding-tightness ordering (augmented-SCM ridge sweep) that reveals whether a gap tracks the control construction rather than the treatment—demonstrated on the two-round tariff design and calibrated on a clean-control positive control (California Proposition 99).

2. **Donor-free descriptive volatility facts** for the two rounds, with three competing interpretations (learning; shock magnitude and surprise; background conditions) stated and left unresolved.

3. **Explicitly non-causal relative accounting**: S&P 500 relative to an overseas-predicted path (BSTS, with its contaminated-covariate assumption disclosed), and a beta decomposition of the sector cross-section showing that raw "which sector fell most" rankings are dominated by market beta. These are descriptions of relative performance, not tariff effects.

What we do not claim. We do not estimate the effect of tariffs on the S&P 500; we do not interpret any SCM/BSTS gap as causal; we do not claim a fear "channel" share; and we do not claim evidence of market learning. The paper's value is in showing precisely why each of these claims would fail, and what can be said instead.

The remainder proceeds as follows. Section 2 lays out the institutional timeline, including the 2026 litigation aftermath. Section 3 describes the data. Section 4 sets out the estimators and the diagnostic uses to which we put them. Section 5 presents the diagnosis. Section 6 distills the diagnostics into a pre-reporting protocol and demonstrates it on a clean-control positive control. Section 7 discusses interpretation and limitations, and Section 8 concludes.

2 Institutional Background

2.1 Two tariff rounds

Table 1 summarizes the key dates. The essential difference is the delivery mechanism: **2018 was gradual**—tariffs arrived in batches over eighteen months and markets priced each step as it came; **2025 was a shock**—a single announcement covering all major partners at once, followed by a crash and an equally fast partial reversal. Our focal events are 2018-07-06 (List 1, the largest single node of round one) and 2025-04-02 (Liberation Day).

2.2 The 2026 legal aftermath

The legal foundation of the 2025 tariffs collapsed within a year of the event we study: the Supreme Court held in *Learning Resources, Inc. v. Trump* that IEEPA does not authorize presidential tariff imposition (Supreme Court of the United States, 2026); the administration immediately pivoted to a Section 122 surcharge (The White House, 2026), which the Court of International Trade invalidated (U.S. Court of International Trade, 2026) before its 150-day statutory expiry, with Section 301 investigations (Office of the United States Trade Representative, 2026) standing by as the third legal basis in twelve months. This relay matters for our interpretation: it is direct evidence that tariff policy in this period was a moving target whose legal durability was unknown at each moment—economic policy uncertainty in the classic sense of Baker et al. (2016)—and hence background condition, not object, of our analysis.

Table 1: Tariff timeline, 2018–2026 (selected events; compiled from Bown and Kolb, 2025 and Federal Register and court records)

Round	Date	Event
2018	Mar 08	Section 232 steel (25%) and aluminum (10%) tariffs signed
2018	Mar 22	Section 301 China memorandum signed
2018	Jul 06	**List 1 effective, 25% ($34B)** — round-1 focal event
2018	Aug 23	List 2 effective, 25% ($16B)
2018	Dec 01	G20 Buenos Aires truce (90 days)
2019	May 10	List 3 raised 10%→25% ($200B)
2019	Sep 01	List 4A effective, 15% ($112B)
2025	Feb 01	China +10%, Canada/Mexico +25% (IEEPA/fentanyl)
2025	Apr 02	**Liberation Day reciprocal tariffs** — round-2 focal event
2025	Apr 09	90-day pause on reciprocals; China rate raised to 125%
2026	Feb 20	*Learning Resources v. Trump*: Supreme Court (6–3, Roberts, C.J.) holds IEEPA does not authorize the tariffs
2026	Feb 24	Government pivots to Section 122 (Trade Act of 1974): 10% global surcharge (statutory cap 15%, sunset 150 days)
2026	May 07	Court of International Trade (2–1) holds Section 122 unlawful; CAFC administrative stay; CBP keeps collecting
2026	Jul 24	Section 122 statutory expiry; 60 Section 301 investigations (initiated Mar 12) take over as successor authority

3 Data

All price data are daily closes from Yahoo Finance via `yfinance`, 2017-01-01 through 2025-12-30. Table 2 lists the panels.

Table 2: Data panels

Panel	Instruments	Role
us_index	^GSPC ^DJI ^IXIC	Treated unit (S&P 500)
sector_etf	XLY … XLRE (11 ETFs)	Candidate donor pool 1
overseas_index	15 developed markets[a]	Candidate donor pool 2 (extended 2026-07-12)
macro	^VIX, GC=F, ^TNX, DX-Y.NYB	Covariates

[a] ^N225, ^GDAXI, ^FTSE, ^FCHI, ^AXJO, ^GSPTSE, ^STOXX50E, ^SSMI, ^N100, VGK, EZU, EWJ, EPP, EWA, EWC; EEM (emerging) and HSI (China-linked) excluded.

Limitations of the data. `yfinance` covers currently listed tickers; sector-ETF survivorship is mild but nonzero. XLC (listed 2018-06) and XLRE (2018-10) lack pre-period data for the 2018 round, so the two-round comparison (Section 5.5) uses the 9 common donors. Sector China-exposure figures (Section 5.13) are PIIE/BEA-based proxies. VIX futures term-structure data are not reliably available from free sources, so all volatility statements use spot VIX. A GDELT news panel (1,416 items over 2025-04-01–15) was collected for an auxiliary sentiment dimension whose finance-specific tone proved weak (contemporaneous correlation with S&P 500 returns +0.06); it is excluded from the analysis and documented in Appendix A only.

4 Methods and Their Diagnostic Uses

The estimators below are standard; what is unusual is the use we put them to. Because the central assumption of every one of them (untreated controls) fails in a known direction, we treat each estimator's output as a *diagnostic statistic* whose behavior under contamination we document, rather than as an effect estimate.

4.1 Synthetic control

Following Abadie et al. (2010), with pre-treatment period T_0 we solve for donor weights w:

$$\min_{w} \sum_{v} V_v \Big(X_{1v} - \sum_{j} w_j X_{jv} \Big)^2 + \big\| Z_1 - Z_0 w \big\|^2 \qquad \text{s.t. } w_j \geq 0, \ \ \textstyle\sum_j w_j = 1, \tag{1}$$

where V_v weights the predictors. We report both equal V and the data-driven choice of Abadie (2021) (`pysyncon`), because Section 5.2 shows the gap is sensitive to it. Weights are estimated with `cvxpy` (CLARABEL) and cross-checked against `pysyncon`. Gaps are computed on cumulative log returns.

4.2 Inference tools read as power diagnostics

The *permutation placebo* (Abadie, 2021) iteratively treats each donor as the treated unit; the permutation p-value is the share of placebo gaps at least as extreme as the realized one. Under donor homogeneity this test has no power by construction, and a p-value of 100% is itself the diagnostic finding. *Leave-one-out* (LOO) re-estimation exposes dependence on any single donor. *In-time placebos* move the intervention date into the pre-period and locate the realized statistic in the resulting null distribution; we report these quantiles as calibration statistics, not as significance tests of an effect.

4.3 Bayesian structural time series as relative accounting

Brodersen et al. (2015) fit a state-space model (local level plus covariate regression) on the pre-period and forecast a counterfactual. We implement the frequency-domain equivalent with `statsmodels UnobservedComponents`. Two assumptions fail here: the covariates (overseas indices) are treated, and—unlike convex SCM—the regression extrapolates without simplex constraints, so contamination passes through coefficients rather than being averaged. We therefore read its output exclusively as *relative-performance accounting*: where the S&P 500 stood relative to an overseas-predicted path over the same window.

4.4 Event study and its limits

Following MacKinlay (1997), the market model $R_{it} = \alpha_i + \beta_i R_{mt} + \varepsilon_{it}$ is estimated over an estimation window and abnormal returns cumulated over the event window. In a market-wide shock, however, the market factor is itself the treated variable and contains each sector's own return; residuals therefore describe *relative* performance, not causal abnormal returns. We use the event study for one descriptive purpose: decomposing the sector cross-section into a beta component and a residual component (Section 5.8).

4.5 Volatility-covariation accounting (not mediation)

For any return series we estimate, over the pre-period,

$$R_t = \alpha + \beta\, \Delta VIX_t + \varepsilon_t, \qquad \Delta VIX_t = \text{VIX log return}, \tag{2}$$

and use it two ways. First, an *orthogonalized SCM*: treated and every donor are purged with their own pre-estimated β (out of sample over the event window), and the SCM re-run on the purified paths. Second, a *descriptive decomposition* of the event-window cumulative return: $\sum_{t\in E} R_t = \hat{\beta} \sum_{t\in E} \Delta VIX_t + \sum_{t\in E} \hat{\varepsilon}_t$, the first term measuring contemporaneous covariation with the volatility state.

Three properties of this exercise must be stated before any number is read. (i) ΔVIX is contemporaneous, and VIX is the implied volatility of S&P 500 options: the return–ΔVIX relation embeds the leverage/volatility-feedback mechanics, so covariation shares are close to re-descriptions of the decline itself. (ii) $\hat{\beta}$ is estimated on a calm pre-period and extrapolated across a regime shift (Section 5.6). (iii) Sequential ignorability, required for causal mediation (Imai et al., 2010), fails outright. We therefore do not use the language of channels, mediation, or causal shares anywhere in the results.

4.6 GARCH and VIX paths (donor-free)

For each round we fit a GARCH(1,1) with an event dummy in the variance equation,

$$\sigma_t^2 = \omega + \alpha\, \varepsilon_{t-1}^2 + \beta\, \sigma_{t-1}^2 + \gamma\, D_t, \tag{3}$$

and, more importantly, read the shape of the VIX path (level change, time to peak, half-life). Neither statistic uses a control group; both are descriptive.

4.7 Augmented SCM as a binding-tightness dial

The convex hull constraint of equation (1) confines the counterfactual to the donor simplex. The augmented SCM of Ben-Michael et al. (2021) (see also Doudchenko and Imbens, 2016) replaces convex weights with ridge regression, anchored at the convex-SCM benchmark and level-anchored at the event date, so results remain comparable to SCM cumlog gaps. We use the ridge penalty sweep $\alpha \in [0.01, 100]$ as a dial that moves the counterfactual continuously from "fully bound to donors" (convex) to "essentially unbound" (level line). If the gap orders monotonically along this dial, it tracks the control construction; the script `ascm_sweep.py` reproduces Table 5.

4.8 Neural-network maps as extrapolation-error probes

MLP and LSTM models map donor returns plus VIX to S&P 500 returns (pre-period training, event-period out-of-sample application; 5 seeds; window lengths 3/5/10). Their outputs are *prediction errors of a nonlinear pre-period map*—structurally contaminated in the same way as SCM—and we report them only to document the magnitude and instability of nonlinear extrapolation error, never as gaps.

5 Results: The Diagnosis

5.1 The Liberation Day window

The reciprocal tariffs were announced after the close on 2025-04-02 (close: 5,671). Over the following four trading sessions (April 3, 4, 7, and 8) the S&P 500 fell to a trough close of 4,983 on April 8 (-12.1%); the April 9 pause triggered a $+9.52\%$ single-day rebound. The sector cross-section ranged from XLE (energy) -18.8% to XLP (staples) -6.4%. Every overseas index in the donor pool fell over the same window (-7.6% to -13.3%). Two properties of this window drive everything that follows: the shock was global (no market was untreated), and the domestic "controls" are components of the treated index.

5.2 Sector SCM: homogeneity by construction

Over the pre-period 2024-06–2025-04, the equal-V SCM loads XLK 0.36, XLY 0.18, XLV 0.17, XLF 0.09 (top four: 79.8%); the pre-trend statistic is $p = 0.957$ (Newey–West)—though, as Section 5.3 shows, this design has no power, so a high pre-trend p-value is uninformative rather than reassuring. Realized gaps: short window (4/2→4/8) −0.04%; long window (4/2→6/30) +0.68%. The near-zero gaps are not evidence that tariffs were harmless; they are what this construction must produce when donors are slices of the treated index—the synthetic control approximately *reconstructs* the S&P 500, weight-fitting details aside. Switching to data-driven V (Abadie, 2021) disperses the weights and moves the long-window gap to +7.45%: an 11× spread from the equal-V benchmark attributable to the weight matrix alone. A statistic that moves by an order of magnitude with a defensible nuisance choice is not measuring the treatment.

5.3 Placebo and leave-one-out: no power by construction

The permutation test returns $p = 100\%$: the realized short-window gap (−0.04%) has the smallest absolute value among all 12 units, against placebo gaps ranging −6.24% to +4.69%. This is the constructive no-power result announced in the introduction: with homogeneous donors, synthetic ≈ treated and the gap is pinned near zero *irrespective of the intervention*; the test cannot reject anything, ever, in this design. One caveat on interpretation: a smallest-of-all realized gap is observationally consistent with two readings—the homogeneity artifact we emphasize, or genuine treatment heterogeneity in which the cap-weighted, globally diversified index is simply the least tariff-exposed unit in the pool while concentrated sector funds absorb larger shocks. Our design cannot separate the two, and nothing downstream depends on choosing between them. LOO gaps span [−0.49%, +0.69%]—stable, for the same reason: every donor subset still reconstructs the index. Stability here is a symptom of the design, not a robustness property of an estimate.

5.4 Overseas controls: SCM unusable, BSTS as relative accounting

With the overseas pool at 15 developed markets, SCM weights concentrate on Canada (0.76) and Australia (0.21) — tickers ^GSPTSE and ^AXJO — and pre-trends fail outright ($p = 0.000$); the long-window statistic (+2.43%) is uninterpretable and we discard it. Extending the pool did not dilute concentration (an earlier pool put 0.99 on EEM): in any overseas family there is always one index that tracked the S&P 500 best pre-event, and convex SCM gives it the weight.

The BSTS regression of the S&P 500 on the equal-weighted overseas basket is retained only as relative accounting: the S&P 500 closed the short window 3.54% below its overseas-predicted path and the long window 4.35% above it (posterior tail-area statistics 60% and 8.2%; in-time placebo quantiles 10% and 60%). These numbers describe where the U.S. market stood relative to a (treated) world basket—a descriptive comparison across contaminated markets, in which the U.S. policy reversal of April 9 plausibly plays a role. They are not tariff effects, and we do not pool them with SCM statistics (Section 5.12).

5.5 Two rounds: realized reactions versus constructed gaps

Table 3 compares the rounds on the 9 common donors. The windows deliberately follow each round's natural rhythm—2018: short = event to +2 months (2018-09-06), long = to year-end; 2025: short = event to +4 sessions (2025-04-08), long = to June 30—because the delivery mechanisms differ; the notebook records this as design, not oversight. Because the realized cells therefore confound shock with window length, we re-ran both rounds at *aligned* horizons of +4, +20, and +60 sessions (`aligned_windows.py`). Three readings follow. (i) At +4 sessions the crash contrast is intact: +1.4% (2018) versus −12.1% (2025). (ii) By +20 sessions it has largely dissipated (+2.9% versus −1.2%), and at +60 both rounds are positive (+6.0% versus

+9.4%): the dramatic long-window contrast in Table 3 (−9.2% versus +9.4%) is manufactured by unequal windows—the 2018 long cell runs six months and absorbs the Q4 2018 drawdown, while at the aligned three-month point (October 1) the 2018 return stood at +6.0%. (iii) In every aligned cell the SCM statistic stays below 1% (range −0.09% to +0.91%), consistent with the construction story of Section 5.2. The price cells—realized or synthetic—are not the carrier of the between-round difference; anything substantive lives in the volatility signature (Section 5.6). Note also that the 2025 short-window statistic here (−0.09%) uses the 9-donor pool and therefore differs slightly from the 11-donor figure of Section 5.2 (−0.04%); both are reported, neither is an effect.

Table 3: Two-round comparison (9 common donors; windows follow each round's rhythm, see text; descriptive—no cell is an effect)

Round	Realized (short / long)	SCM statistic (short / long)	MSPE	Pre-trend
2018–19 (List 1)	+4.3% / −9.2%	+0.03% / −0.69%	7.2	$p = 0.727$
2025 (Liberation Day)	−12.1% / +9.4%	−0.09% / +0.91%	4.7	$p = 0.775$

5.6 Volatility signatures: the donor-free facts

These statistics use no control group and are therefore the only findings in this paper that do not inherit contamination; they are, accordingly, descriptive statements about realized paths (Figure 1):

- **2025 Liberation Day:** VIX closed at 21.8 on the eve of the announcement (April 1) and peaked at 52.3 on April 8, four sessions into the crash (+140%), then a half-life of about one day—the April 9 pause unwound the spike almost immediately.
- **2018 List 1 (Jul 06):** VIX closed at 15.0 on the eve of the node (July 5) and never exceeded that level over the following 60 sessions (window peak 14.9)—no immediate reaction at the node; VIX climbed only as nodes accumulated into Q4.

The GARCH(1,1) fits are secondary: both rounds show high persistence ($\alpha + \beta > 0.96$) over ±120-day windows, and the event-dummy regression is coarse (γ_{18} significantly negative, γ_{25} insignificant—an ordering built on one insignificant coefficient is not worth interpreting). We rest nothing on the parameters; the path shapes carry the content. Three interpretations of the contrast are live—market learning across rounds; larger and more surprising treatment in 2025; different rate/inflation/AI-cycle backgrounds—and our design cannot separate them (Section 7.3).

5.7 Volatility-covariation accounting

Applying equation (2): the treated pre-period coefficient is $\hat{\beta} = -0.079$. The orthogonalized SCM (treated and all donors purged, both with pre-estimated coefficients) moves the statistics only slightly: short −0.04% → −0.09%, long +0.68% → +0.86% (Table 4)—the contemporaneous covariation largely cancels in the treated-minus-synthetic difference because it hits both sides. The descriptive decomposition of the event-window return attributes a −6.85-point fitted covariation component to the short-window cumulative move of −12.27% (residual component −5.41; a covariation share of roughly 56%; long window +0.79 of +7.64, residual +6.85). Given the three properties stated in Section 4.5—mechanical contemporaneous link, regime extrapolation of $\hat{\beta}$, failed ignorability—this share should be read as a descriptive statistic about how tightly the decline co-moved with the options market's repricing of volatility, nothing more. It is not a channel, not causal, and its precision is illusory; we flag that any study regressing index returns on their own implied volatility will "recover" a large share by construction.

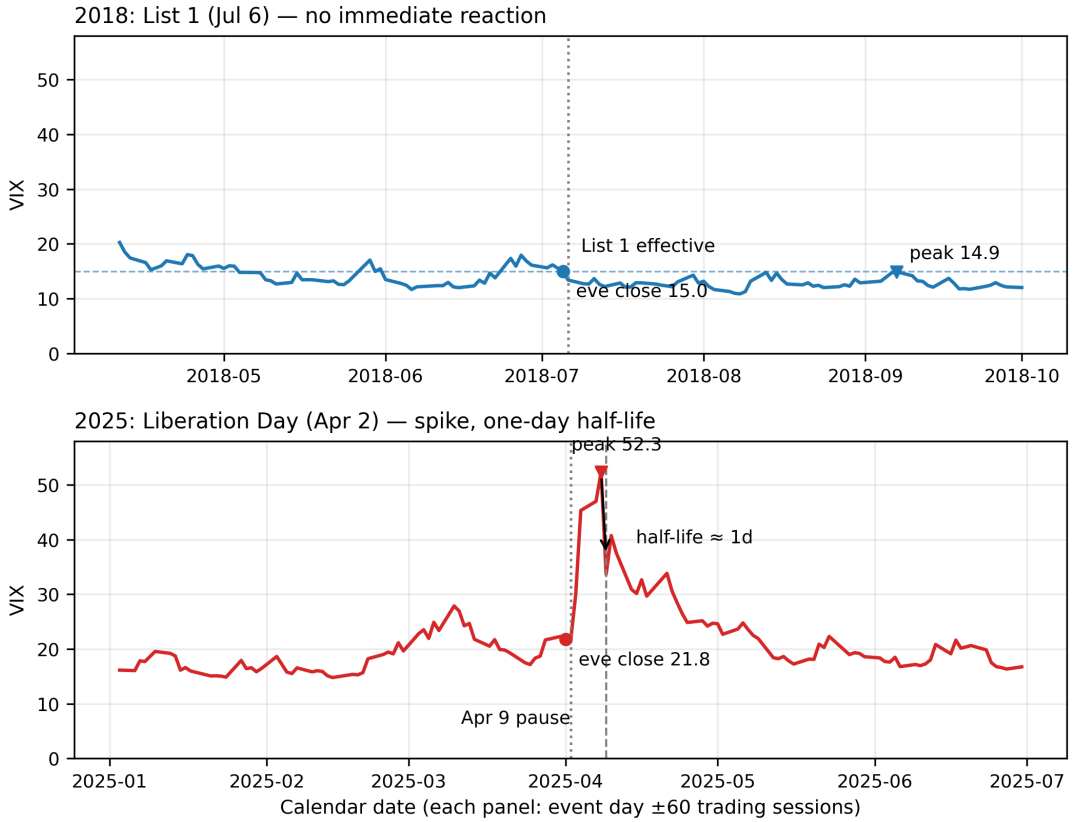


Figure 1: VIX around each focal node, ±60 trading sessions, calendar axes (a shared relative-event axis would visually compress 2018's null reaction against 2025's spike). Upper: List 1 (2018-07-06)—the eve close (15.0) is never exceeded afterward (window peak 14.9). Lower: Liberation Day (2025-04-02)—eve close 21.8, peak 52.3 on April 8 (+140%), half-life about one day after the April 9 pause. Generated by `make_vix_figure.py`.

Table 4: SCM statistic before and after VIX-orthogonalization (both treated and donors purged with pre-period coefficients)

Window	Raw SCM statistic	Purified	Difference
Short (4/2→4/8)	−0.04%	−0.09%	−0.05pp
Long (4/2→6/30)	+0.68%	+0.86%	+0.18pp

5.8 Sector cross-section: beta dominates raw rankings

Over 4/2→4/8 the raw sector ranking (XLE −18.8% worst) and the market-model residual ranking differ sharply: on residuals, XLE stands at −15.16 points and XLK at +5.19 points, while XLK's raw return was the worst (−14.22%) against a fitted β of 1.59. The cross-sectional correlation between raw returns and residual ranks is only 0.24: raw "which sector fell most" league tables are dominated by market beta rather than sector-specific exposure. We stop the claim there. Because the market factor is the treated index and contains the sectors themselves, residuals cannot be read as causal "abnormal" performance; the +5.19-point residual for XLK reflects both its genuine relative strength during the rebound and the mechanical absorption of its own weight in the market factor. The usable output is the descriptive decomposition, not a label of winners and losers.

5.9 The XLK mechanism

If the near-zero sector-SCM statistic is a homogeneity artifact, removing the dominant homogeneous donor should break it. It does: excluding XLK moves the long-window statistic from +0.68% to +7.51% (XLK's contribution −6.83 points; post-exclusion weights XLY 0.48, XLV 0.31, then XLB/XLE; pre-trend $p = 0.207$). The mechanism is transparent: with XLK in the pool the synthetic control inherits the post-event tech-led rebound and tracks the treated unit; without XLK it misses that rally. The data-driven-V result of Section 5.2 (+7.45%) works through the same channel—both stop matching the pre-period AI rally with XLK—so the two numbers are one mechanism observed twice, not independent corroboration.

5.10 Augmented SCM: gaps order with binding tightness

Table 5 reports the ridge sweep. As $\alpha \to 0$ the ridge approaches unconstrained donor fitting and the statistic approaches the convex SCM; as $\alpha \to \infty$ coefficients shrink to zero, the "counterfactual" degenerates to an anchored level line, and the statistic converges to the raw return itself ($\alpha = 100$: −11.88% vs. realized −12.1% short; +9.40% vs. +9.4% long). At $\alpha = 1$ the ridge coefficients sum to 0.574 (vs. 1.0 under convexity): the construction is part donor, part extrapolation.

Table 5: Augmented SCM: statistic by ridge penalty α (cumlog metric, event-date anchored; `ascm_sweep.py`)

Window	SCM (convex)	α=0.01	α=0.1	α=1	α=10	α=100
Short (4/2→4/8)	−0.04%	−1.13%	−2.09%	−5.06%	−9.56%	−11.88%
Long (4/2→6/30)	+0.68%	+0.32%	+1.37%	+4.19%	+7.75%	+9.40%

The monotone ordering is the finding. The statistic sweeps continuously from the convex bound to the raw return as the dial loosens the donors' grip: it is a function of the control construction, and there is no setting of α at which it can be claimed as a treatment effect. The near-zero convex value is not "tariffs were harmless"; it is "under the tightest binding the design cannot see the shock."

5.11 Nonlinear extrapolation error

The neural maps produce large, unstable numbers: MLP +10.29%±7.02pp and LSTM +16.18%± 9.39pp across 5 seeds (a linear map gives +0.16%); window sensitivity runs +15%, +18%, and +21% for $w = 3$, 5, 10, and single seeds reach +35%. Validation/train loss ratios near 1.2 over a more volatile validation segment confirm overfitting. We draw no inference from these magnitudes beyond what they demonstrate: a nonlinear pre-period map extrapolated into a

regime shift yields errors two orders of magnitude above the linear map's, unstable across seeds. These are prediction errors, not counterfactual gaps, and they are excluded from Table 6.

5.12 Specification map

Table 6 collects the long-window statistics grouped by what each actually measures. The grouping is the point: rows within the first block are one nominal estimand fractured by construction choices; the second block is a different estimand (relative to a treated world basket); the third block is not a gap at all. Presenting such a collection as an "uncertainty interval" over one quantity would be a category error.

Table 6: Specification map: long-window statistics by estimand. No row is a tariff effect; within-block spread orders with donor-binding tightness (Sections 5.2–5.10).

Estimand	Specification	Long-window statistic
S&P minus sector-synthetic	Convex SCM, equal V	+0.68%
	Convex SCM, data-driven V	+7.45%
	Exclude XLK	+7.51%
	VIX-purified (equal V)	+0.86%
	Linear donor map	+0.16%
	Augmented SCM, $\alpha \in [0.01, 100]$	+0.32% to +9.40%
S&P minus overseas-predicted path	Overseas SCM	+2.43% (discarded, pre-trend failure)
	BSTS regression	+4.35% (relative accounting)
Extrapolation error (not a gap)	MLP, 5 seeds	$+10.29\% \pm 7.02$pp
	LSTM, 5 seeds	$+16.18\% \pm 9.39$pp
Reference	Realized return	+9.4%

5.13 Reading the SCM weights: China exposure

Why did the equal-V SCM load 36% on XLK? Not tariff exposure: XLK's China-exposure proxy (∼20%) exceeds the index's (∼7.5%), and the weighted synthetic exposure (∼11.3%) is *higher* than the actual index's. The weights reflect pre-period path coincidence (the AI rally), which is exactly the homogeneity mechanism of Section 5.9. The exposure lens clarifies the weights' economic content and nothing about any gap.

6 A Pre-Reporting Diagnostic Protocol, with a Positive Control

The tariff analysis is a failure case: every diagnostic fired. For the diagnostics to be a protocol rather than a post-mortem, a reader must also see what *passing* looks like. This section states the protocol and runs it end-to-end on the canonical clean-control application—California Proposition 99 in Abadie et al. (2010)—as a positive control.

6.1 The checklist

Before any synthetic-control gap is reported as an effect for a market-wide or cross-sectional shock:

1. **Donor–treatment homogeneity.** Compute the pre-period correlation between the treated unit's path and each donor's path (returns or first differences, not levels). If donors are slices of the treated unit or correlations are extreme, the synthetic control will

reconstruct the treated series and the gap is pinned near zero by construction. *Tariff case:* max correlation 0.91 (XLK), donors are index components ⇒ fail (Sections 5.2, 5.9).

2. **Placebo power.** Run permutation placebos and ask whether the realized gap is *extreme* among them. If the realized gap is the smallest in absolute value, the test has no power against any alternative, and neither significance nor insignificance is interpretable. *Tariff case:* $p = 100\%$, smallest of 12 units ⇒ fail (Section 5.3).

3. **Weight concentration and pre-trends.** Inspect the weight vector (nonzero count, top share) and the pre-trend statistic. Single-donor domination and pre-trend failure disqualify the construction. *Tariff case:* overseas pool top weight 0.76 (Canada), pre-trend $p = 0.000$ ⇒ fail (Section 5.4); note that a *high* pre-trend p-value is equally uninformative under step-2 failure.

4. **Binding-tightness sweep.** Sweep a regularization dial (augmented-SCM ridge penalty) from fully donor-bound to unbound. If the gap orders monotonically along the dial—from the convex bound to the raw return—it tracks the control construction, not the treatment. *Tariff case:* $+0.68\% \to +9.40\%$ monotone in α ⇒ fail (Section 5.10).

A gap may be reported as an effect only when step 2 shows power and step 4 is flat. The tariff events fail all four steps in known directions, which is why Section 5 reports a diagnosis rather than estimates.

6.2 Positive control: California Proposition 99

We apply the same four diagnostics to the classic clean-control case: California's 1988 tobacco program, with 38 untreated donor states and annual cigarette sales per capita, 1970–2000 (`positive_control.py`; the ADH predictor set via `pysyncon`; data mirrored alongside the script). This is an environment the protocol is supposed to pass: donors are other states, not slices of California, and a large published effect exists against which the diagnostics can be calibrated. Table 7 reports all four steps side by side with the tariff case.

The convex SCM recovers the published effect: California's 2000 gap is -25.5 packs per capita, within the range reported by Abadie et al. (2010) (roughly -18 to -25 across specifications). Around that estimate the diagnostics behave as a protocol should:

- **Homogeneity.** State-level co-movement is not trivial (max first-difference correlation 0.84, mean 0.58), so high correlation *alone* is not disqualifying. What the tariff case has and this case lacks is *containment*: no donor state is a component of California, whereas XLK (correlation 0.91) is a component of the treated index, and the synthetic control reconstructs it.

- **Placebo power.** On the naive terminal-gap statistic the test is still weak (California ranks 7th of 39, $p = 0.179$)—a useful reminder that raw terminal gaps are noisy even in the canonical case. On the ADH-style post/pre RMSPE ratio, California is the single most extreme unit of 39 (ratio 9.2, permutation $p = 0.026$). The contrast with the tariff case is photographic: there the realized gap is the *smallest* of all units ($p = 1.00$).

- **Concentration.** Weights spread over 9 of 38 donors (top-1 0.31, top-3 0.66), versus 0.76 on a single index in the tariff overseas pool.

- **Binding tightness.** The effect exists under the tightest binding (convex: -25.5); the ridge dial varies its magnitude within the same regime (-19.5 to -33.8 over $\alpha \in [0.01, 100]$) without approaching the raw change (-48.5) or collapsing to zero. Here the dial adjusts the estimate; in the tariff case the dial *was* the estimate.

Table 7: The four diagnostics: tariff events (failure case) versus California Proposition 99 (positive control). Positive-control numbers from `positive_control.py`.

Diagnostic	2018/2025 tariff events	California Prop. 99
Donor homogeneity	max corr 0.91; donors are components of the treated index	max corr 0.84, no containment (donors are other states; mean 0.58)
Placebo power	realized gap smallest of 12 units, $p = 1.00$	RMSPE-ratio most extreme of 39 units, $p = 0.026$ (terminal-gap $p = 0.179$)
Weight concentration	overseas top-1 0.76 (Canada)	9 of 38 nonzero, top-1 0.31, top-3 0.66
Binding-tightness sweep	monotone: convex $+0.68\% \rightarrow$ raw $+9.40\%$	effect present at every α (-19.5 to -33.8); convex -25.5; raw change -48.5

7 Discussion

7.1 One contaminated world, many fractured estimands

The classical SCM application (California tobacco vs. other states) assumes untreated donors. A global tariff shock violates this for every financial asset simultaneously, in the same direction: all controls fall with the treated unit, pulling every synthetic path down and compressing every treated-minus-synthetic statistic toward zero relative to whatever the unobservable counterfactual would have been. The dispersion across specifications in Table 6 measures how differently the various constructions propagate that contamination—it is not uncertainty about one quantity. A synthetic-control statistic in this setting is best understood as a well-defined linear functional of (treated, donors) paths whose causal content is nil.

7.2 Why market adjustment cannot isolate sector effects here

In firm-level event studies the market model removes common noise because the event is firm-specific. In a market-wide treatment the market factor is the treatment, and each sector's own return sits inside it (XLK is the largest weight in the S&P 500). Residuals therefore mix genuine relative strength with mechanical own-weight absorption. Cross-sectional statements of the form "sector X was abnormally resilient" are unavailable in this design; what is available is the descriptive decomposition of rankings into beta and residual components.

7.3 Two rounds, three interpretations

The volatility contrast (immediate spike, one-day half-life in 2025; no immediate reaction in 2018) admits at least three readings: the market learned the escalation template between rounds; the 2025 treatment was larger and more surprising than any single 2018 node; the background (rates, inflation, AI cycle) differed. Treatment intensity differs by an order of magnitude across the focal dates, so nothing identifies the learning channel. We present the contrast as fact and the interpretations as open.

7.4 Limitations

1. **No clean control**: every candidate donor is treated; this is the paper's subject and its boundary.

2. **Survivorship**: `yfinance` covers listed tickers; mild for sector ETFs, nonzero.

3. **2018 simplification**: one focal node represents a seven-node escalation.

4. **VIX mechanical identity**: any return-on-ΔVIX exercise is contemporaneously mechanical; covariation shares are descriptive only, and $\hat{\beta}$ is extrapolated across a regime shift.

5. V–w **coupling**: the data-driven predictor weights are chosen on pre-period outcomes and then interact with the weight estimation; optimizing pre-fit can worsen post-period validity. This known SCM critique is visible here as the 11× spread of Section 5.2.

6. **Estimand discipline**: no statistic in Table 6 is an effect; we have flagged each block's actual content.

7. **China exposure is a proxy** (PIIE/BEA synthesis), used only to interpret weights.

8. **Learning not identified**: treatment magnitude, surprise, and background all differ across rounds.

8 Conclusion

This paper asked what counterfactual methods deliver when a policy shock treats every candidate control, and answered with a diagnosis. Four findings define the limits. Placebo tests have no power when donors are homogeneous slices of the treated unit; convex SCM concentrates overseas weights on a single best tracker and fails pre-trends; counterfactual statistics order monotonically with a binding-tightness dial (the augmented-SCM ridge sweep) that has nothing to do with the treatment; and the apparent spread of "estimates" across methods is a collection of different estimands, none causal. A mediation-style decomposition of the crash through VIX was shown to be mechanically flawed—VIX is the implied volatility of the treated index's own options—and was retained only as descriptive covariation accounting.

The firm empirical content that survives is donor-free and descriptive: around the 2025 Liberation Day announcement, VIX rose about 140% from its pre-announcement close to its April 8 peak and decayed with a one-day half-life after the April 9 pause, while the comparable 2018 node produced no immediate VIX reaction; and the sector cross-section's raw decline rankings are dominated by market beta. The 2026 legal collapse of the tariff authorities (IEEPA struck down; Section 122 expired under injunction; Section 301 pending) underlines the environment these markets were pricing: policy as a moving target.

For practitioners, Section 6 distills the diagnosis into a four-step pre-reporting protocol—homogeneity, placebo power, weight concentration, binding-tightness sweep—calibrated on a clean-control positive control. A gap may be reported as an effect only when placebos have power and the statistic is invariant to the dial. In the two tariff rounds studied here—and, we suspect, in many applications to synchronized market shocks—those conditions fail, and the responsible output is the diagnosis itself.

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A Reproducibility

All results are generated by fifteen Jupyter notebooks, each produced by a `generate_0X_*.py` script and sharing the library `paper_lib.py` (SCM and augmented SCM via `cvxpy`, BSTS via `statsmodels`, event study, covariation accounting, placebo/LOO, GARCH via `arch`). Data are pulled by `download_tariff_data.py` (with `download_overseas_standalone.py` as a rate-limit

fallback). The augmented-SCM sweep of Table 5 is reproduced by `python ascm_sweep.py`; the positive control of Section 6.2 by `python positive_control.py` (California panel mirrored as `smoking_california.csv`; the 39 placebo fits run once and are cached, so re-runs are instant); the aligned-horizon comparison of Section 5.5 by `python aligned_windows.py`. An auxiliary GDELT news panel (1,416 items, 2025-04-01–15; notebook `nlp_01`) was explored for a sentiment dimension; its finance-specific tone proved weak and it is not part of the analysis. Table 8 maps sections to notebooks.

Table 8: Notebook map

Notebook	Section
01–02 data exploration / event EDA	5.1
03 sector SCM + V sensitivity	5.2
04 CausalImpact + in-time placebo	5.4
05 event study residuals	5.8
06 placebo / LOO	5.3
07 overseas SCM (discarded)	5.4
08 two-round comparison	5.5
09 retrospective	5 (overview)
10 VIX orthogonalization	5.7
11 GARCH regimes	5.6
12 MLP/LSTM extrapolation errors	5.11
13 covariation decomposition	5.7
14 China exposure	5.13
15 exclude-XLK mechanism	5.9